

Image Analysis Report

Generated	2026-04-24 07:16:12Z
Source	corrosion.jpg
Report ID	00f31dc1-2467-4853-80d5-25bb6b52f9f5



Executive Summary

The assessment confirms a severe level of corrosion, characterized by deep localized scaling and preferential attack along a horizontal weld seam. Immediate NDT is required to quantify metal loss and determine the remaining structural integrity of the component.

Key Findings

#	Finding
1	Heavy localized scaling and delamination of oxidized material in the central region of the image.
2	Corrosion is significantly concentrated along a horizontal linear feature, likely a weld seam or lap joint.
3	Extensive rust staining spreading vertically from the primary corrosion sites, indicating moisture runoff and active oxidation.
4	Visible failure of the original coating system with large areas of exposed and oxidized substrate.
5	Localized deep pitting is highly probable beneath the heavy scale buildup.
6	Actual remaining wall thickness and internal condition cannot be determined from this visual inspection alone.

Localized detections (VLM, approximate)

- Severe scaling (conf. 95%): x=0.340, y=0.150, w=0.320, h=0.680
- Linear corrosion (conf. 89%): x=0.000, y=0.420, w=1.000, h=0.150
- Coating failure (conf. 85%): x=0.000, y=0.550, w=0.300, h=0.400

Recommendations

#	Recommendation
1	Perform Ultrasonic Thickness (UT) measurements at the center and along the horizontal seam within 72 hours to determine remaining wall thickness.
2	Execute abrasive blasting to SSPC-SP 10 (Near-White Metal) standards to remove all corrosion products for a detailed surface inspection.
3	Evaluate the need for weld repair or the installation of a reinforcement sleeve if wall loss exceeds the corrosion allowance.
4	Conduct a chloride test on the substrate after cleaning to ensure no salts are trapped, which could cause premature coating failure.
5	Apply a high-performance coating system consisting of a zinc-rich primer and a high-build epoxy mid-coat.
6	Implement a routine inspection program using NDT for the entire length of this joint to monitor for similar preferential corrosion.